

Thm. Every non-constant analytic function is open map.

Proof. Suppose that  $f: D \rightarrow \mathbb{C}$  is analytic on the connected open set  $D$ .

Let  $U \subseteq \mathbb{C}$  be an open set. Given  $z_0 \in U$ , choose  $\delta > 0$  such that  $\overline{B}_\delta(z_0) \subseteq U$  and  $f(z) - f(z_0) \neq 0$  for all  $|z - z_0| = \delta$ . (It is possible by the identity theorem.)

Choose  $m = \min\{|f(z) - f(z_0)| \mid |z - z_0| = \delta\}$ , by the min-max theorem.

Let  $w_1 \in \{w \in \mathbb{C} \mid |w - f(z_0)| < m\}$ . Then, on  $|z - z_0| = \delta$ ,

$$|w_1 - f(z_0)| < m \leq |f(z) - f(z_0)|.$$

By the Rouché's theorem,

$$[f(z) - f(z_0)] + [f(z_0) - w_1] = f(z) - w_1$$

has same zeros in  $|z - z_0| < \delta$  to  $f(z) - f(z_0)$ . So,  $f(z) - w_1 = 0$  has at least one zero, that is,  $f(z) = w_1$  at least once in  $|z - z_0| < \delta$ .

Since  $w_1$  was arbitrary,  $\{w \in \mathbb{C} \mid |w - f(z_0)| < m\} \subseteq f[B_\delta(z_0)]$ .